

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS EXTENSION II

HSC Assessment TASK II

JUNE 2004

Time allowed: 70 minutes

Instructions:

- Show all necessary working in every question.
- Start each question on a new page.
- Attempt all questions.
- All questions are of equal value.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- This test forms part of your HSC assessment.
- These questions are to be handed in with your answers.
- Standard integrals are attached at the back of this paper.

Name: _____

Question 1	Question 2	Question 3	Total

Question 1

a) Find $\int \frac{dx}{\sqrt{x^2 - 2x + 5}}$ 2

b) $(x+1)^2$ is a factor of $P(x) = x^5 + 2x^3 + ax + b$. 3
Find a and b .

(c) The real roots of $P(x) = x^3 + 4x - m$ are α, β, γ . 5

i) Find the value of $\frac{1}{\alpha^2 \beta \gamma} + \frac{1}{\beta^2 \alpha \gamma} + \frac{1}{\gamma^2 \alpha \beta}$

ii) Explain why $\frac{1}{\alpha^2 \beta \gamma} = \frac{1}{m \alpha}$

iii) Hence or otherwise form a cubic polynomial whose roots are

$$\frac{1}{\alpha^2 \beta \gamma}, \frac{1}{\beta^2 \alpha \gamma} \text{ and } \frac{1}{\gamma^2 \alpha \beta}.$$

(d) Evaluate $\int_{\pi/6}^{\pi/2} \frac{\cos^3 x}{\sin^2 x} dx$ 3

(e) i) The point $P(4 \sec \theta, 5 \tan \theta)$ lies on a hyperbola. Find the equation of the hyperbola. 4

ii) Find an expression for the gradient of the tangent at the point P as a single trigonometric ratio.

Question 2

a) Evaluate $\int_0^2 \sqrt{4-x^2} \, dx$ 1

b) i) Find real numbers a and b such that 4

$$\frac{2x^2 + 3x + 5}{(x-1)(x^2 + 4)} = \frac{a}{x-1} + \frac{b}{x^2 + 4}$$

ii) Hence find $\int \frac{2x^2 + 3x + 5}{(x-1)(x^2 + 4)} dx$

c) The point $Q \left(ct, \frac{c}{t} \right)$ in the first quadrant lies on the hyperbola $xy = c^2$. 6

i) What are the possible values of the parameter t ?

ii) The line $y = \frac{c}{t}$ cuts the major axis of the hyperbola at T . Find the co-ordinates of M , the mid point of QT .

iii) Find the locus of M giving any restrictions.

d) i) Find the general solution of $\cos 5\theta = 1$. 6

ii) Using the result $\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$, solve the equation $16x^5 - 20x^3 + 5x - 1 = 0$.

ii) Hence show without using direct calculation that

$$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}.$$

Question 3

- a) $P(x)$ is an even polynomial with real co-efficients having the following properties: 2
- α) it has a complex zero
- β) $P(1) = 2$ and $P(2) = -1$.
- State the least degree of $P(x)$ giving reasons.
- b) By using integration by parts or otherwise find $\int \cos^{-1} x \, dx$ 3
- c) Find $\int \frac{du}{\sqrt{u} + 1}$ 3
- d) i) Show that the equation of the normal to the hyperbola $xy = 4$ 5
- at $(2t, \frac{2}{t})$ is $t^3x - ty + 2(1 - t^4) = 0$.
- ii) Find all the possible values of t if the normal passes through the point $(16, \frac{1}{4})$.
- e) The polynomial $P(x)$ is given by $P(x) = x^5 - 5cx + 1$ where c is a real number. 4
- By considering the turning points of $P(x)$, or otherwise, show that $P(x)$ has three distinct real roots if $c > (\frac{1}{4})^{\frac{4}{5}}$

End of Exam

$$\int \frac{dx}{x^2-2x+5} = \int \frac{dx}{(x-1)^2+4}$$

$$= \frac{1}{4} \int \frac{dx}{(x-1)^2+4}$$

$$= \frac{1}{4} \ln \left\{ (x-1) + \sqrt{x^2-2x+5} \right\} + C$$

b) $P(x) = x^5 + 2x^3 + ax + b$

$$P(1) = -1 + 2 - a + b = 0 \quad (1)$$

$$P'(x) = 5x^4 + 6x^2 + a$$

$$P'(1) = 0 \text{ root (multiplicity 2)} = 0 \quad (2)$$

$$5 + 6 + a = 0$$

From (2) $a = -11$

From (1) $b = -8$

c) i) $\frac{1}{x^2\beta\gamma} + \frac{1}{\beta^2\alpha\gamma} + \frac{1}{\gamma^2\alpha\beta} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha^2\beta^2\gamma^2}$

$$= \frac{4}{m^2}$$

ii) $\frac{1}{\alpha^2\beta\gamma} = \frac{1}{\alpha(\alpha\beta\gamma)}$

$$= \frac{1}{\alpha m} \text{ since } \alpha\beta\gamma = m$$

iii) Let $x = \frac{1}{\alpha m}$

$$\therefore \alpha = \frac{1}{mx}$$

Since α is a root of $P(x) = x^5 + 4x - m$

$$\text{then } P(\alpha) = \left(\frac{1}{mx}\right)^5 + \frac{4}{mx} - m$$

$$= 1 + 4m^4x^4 - m^4x^5$$

$$\therefore P(\alpha) = 1 + 4m^4x^4 - m^4x^5$$

$$\int_{\pi/6}^{\pi/3} \frac{dx}{\sin^2 x}$$

Let $u = \sin x$
 $du = \cos x dx$
 $x = \pi/2$
 $x = \pi/3$
 $u = 1$
 $u = 1/2$

$$= \int_{1/2}^1 \frac{1-u^2}{u^2} du$$

$$= \int_{1/2}^1 \frac{1}{u^2} - 1 du$$

$$= \left[-u^{-1} + u \right]_{1/2}^1$$

$$= -2 - [-2 + 1/2]$$

$$= 0.5$$

e) i) $x = 4 \sec \theta, y = 5 \tan \theta$

$$\therefore \frac{x^2}{16} - \frac{y^2}{25} = 1$$

ii) $y = 4x \csc \theta$
 $\frac{dy}{dx} = 4 \sec \theta \tan \theta$
 $y = 5 \tan \theta$
 $\frac{dy}{dx} = 5 \sec^2 \theta$

$$\therefore \frac{dy}{dx} = \frac{5 \sec^2 \theta}{4 \sec \theta \tan \theta}$$

$$= \frac{5 \sec \theta}{4 \tan \theta}$$

$$= \frac{5 \csc \theta}{4}$$

3) complex roots exist in conjugate pairs
 (ii) There is at least one zero between 1 and 2

∴ zero between -1 and -2 because $P(x)$ is even

$$b) \int \cos^{-1} x \, dx = \int \cos^{-1} x \frac{d(\cos^{-1} x)}{dx} dx$$

$$= \cos^{-1} x \cdot x - \int x \cdot \frac{-1}{\sqrt{1-x^2}} dx$$

$$= x \cos^{-1} x - \sqrt{1-x^2} + c$$

$$c) \int \frac{dx}{\sqrt{x+1}}$$

$$\text{Let } u = x+1$$

$$\therefore du = dx$$

$$\int \frac{2x dx}{x+1} = \int \frac{2(x+1)-2}{x+1} dx$$

$$= \int 2 - \frac{2}{x+1} dx$$

$$= 2x - 2 \ln(x+1) + c$$

$$= 2\sqrt{x} - 2 \ln(\sqrt{x}+1) + c$$

$$d) \quad xy = 4 \Rightarrow y = \frac{4}{x}$$

$$\frac{dy}{dx} = -\frac{4}{x^2}$$

$$\text{at } (2^t, \frac{2}{t})$$

$$= \frac{-4}{4t^2}$$

$$= -\frac{1}{t^2}$$

$$\ln H = t^2$$

$$\text{E.g. normal } y - \frac{2}{t} = t^2(x-2t)$$

$$ty - 2 = t^3(x-2t)$$

$$t^3x^2 - ty + 2(1-t^4) = 0$$

$$16t^3 - t + 2(1-t^4) = 0$$

$$64t^3 - t + 8(1-t^4) = 0$$

$$64t^3 - 8t^4 + 8 - t = 0$$

$$8t^3(8-t) + 1(8-t) = 0$$

$$(8-t)(8t^3+1) = 0$$

$$\therefore t = 8 \text{ or } -1/2$$

$$e) \quad P(x) = x^5 - 5cx + 1$$

$$P'(x) = 5x^4 - 5c$$

$$\text{For stationary pt } P'(x) = 0$$

$$\therefore 5x^4 - 5c = 0$$

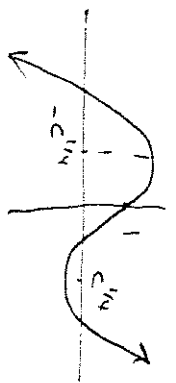
$$5(x^4 - c) = 0$$

$$5(x^2 - c^{1/2})(x^2 + c^{1/2}) = 0$$

$$5(x - c^{1/4})(x + c^{1/4})(x^2 + c^{1/2}) = 0$$

$$\therefore x = c^{1/4} \text{ or } -c^{1/4}$$

$$\therefore \text{Stationary pt at } (c^{1/4}, 1-4c^{5/4}) \quad (-c^{1/4}, 1+4c^{5/4})$$



$$\text{For 3 roots } 1-4c^{5/4} < 0$$

$$-4c^{5/4} < -1$$

$$c^{5/4} > 1/4$$

$$\therefore c^{5/4} < -1/4 \text{ or } c^{5/4} > 1/4$$

∴ does not apply maximum is below x-axis

$$\text{For } c^{5/4} > 1/4$$

$$c > (1/4)^{4/5}$$

2. a)



$$\int_0^2 \sqrt{4-x^2} dx = \frac{1}{4} \pi \times 2^2 = \frac{\pi}{2}$$

b) i) $\frac{2x^2+3x+5}{(x-1)(x^2+4)} = \frac{a(x^2+4)+b(x-1)}{(x-1)(x^2+4)}$

$$= \frac{ax^2+bx+4a-b}{(x-1)(x^2+4)}$$

equating coefficients

$$a=2$$

$$b=3$$

ii) $\int \frac{2x^2+3x+5}{(x-1)(x^2+4)} dx = \int \frac{2}{x-1} dx + \int \frac{3}{x^2+4} dx$
 $= 2 \ln(x-1) + \frac{3}{2} \tan^{-1} \frac{x}{2} + c$

c) i) $t > 0$
 ii) when $y=x \therefore T\left(\frac{c}{t}, \frac{c}{t}\right)$

$$\therefore H = \left(\frac{ct+c}{t}, \frac{c+c}{t} \right)$$

$$= \left(\frac{ct^2+c}{t}, \frac{c}{t} \right)$$

iii) Now $y = \frac{c}{t} \rightarrow t = \frac{c}{y}$

$$x = c \left(\frac{c^2}{y^2} \right) + c$$

$$= \frac{c^3+cy^2}{y^2}$$

$$= \frac{c^2+cy^2}{y^2} \times \frac{y}{2c}$$

$$= \frac{c^2+y^2}{2y}$$

$$\therefore 2xy - y^2 = c^2 \quad \text{for } x, y > 0$$

$$\therefore 16x^5 - 20x^3 + 5x = 1$$

Solution $\cos 5\theta = 1$

$$5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi, \dots$$

$$\theta = 0, 2\pi/5, 4\pi/5, 6\pi/5, 8\pi/5$$

$$\therefore 16x^5 - 20x^3 + 5x - 1 = 0 \quad \text{has 5 solutions}$$

ii) Now

$$\cos 0 + \cos 2\pi/5 + \cos 4\pi/5 + \cos 6\pi/5 + \cos 8\pi/5 = 0$$

but $\cos \theta = 1$

$$\text{and } \cos 8\pi/5 = \cos 2\pi/5$$

$$\cos 6\pi/5 = \cos 4\pi/5$$

$$\therefore 1 + 2\cos 2\pi/5 + 2\cos 4\pi/5 = 0$$

$$2(\cos 2\pi/5 + \cos 4\pi/5) = -1$$

$$\therefore \cos 2\pi/5 + \cos 4\pi/5 = -1/2$$

